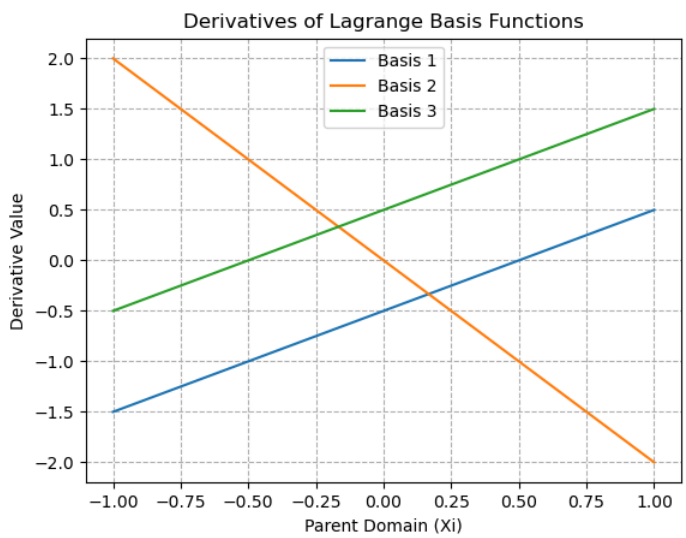
Derivatives of Lagrange Basis Functions

Problems 1 and 2

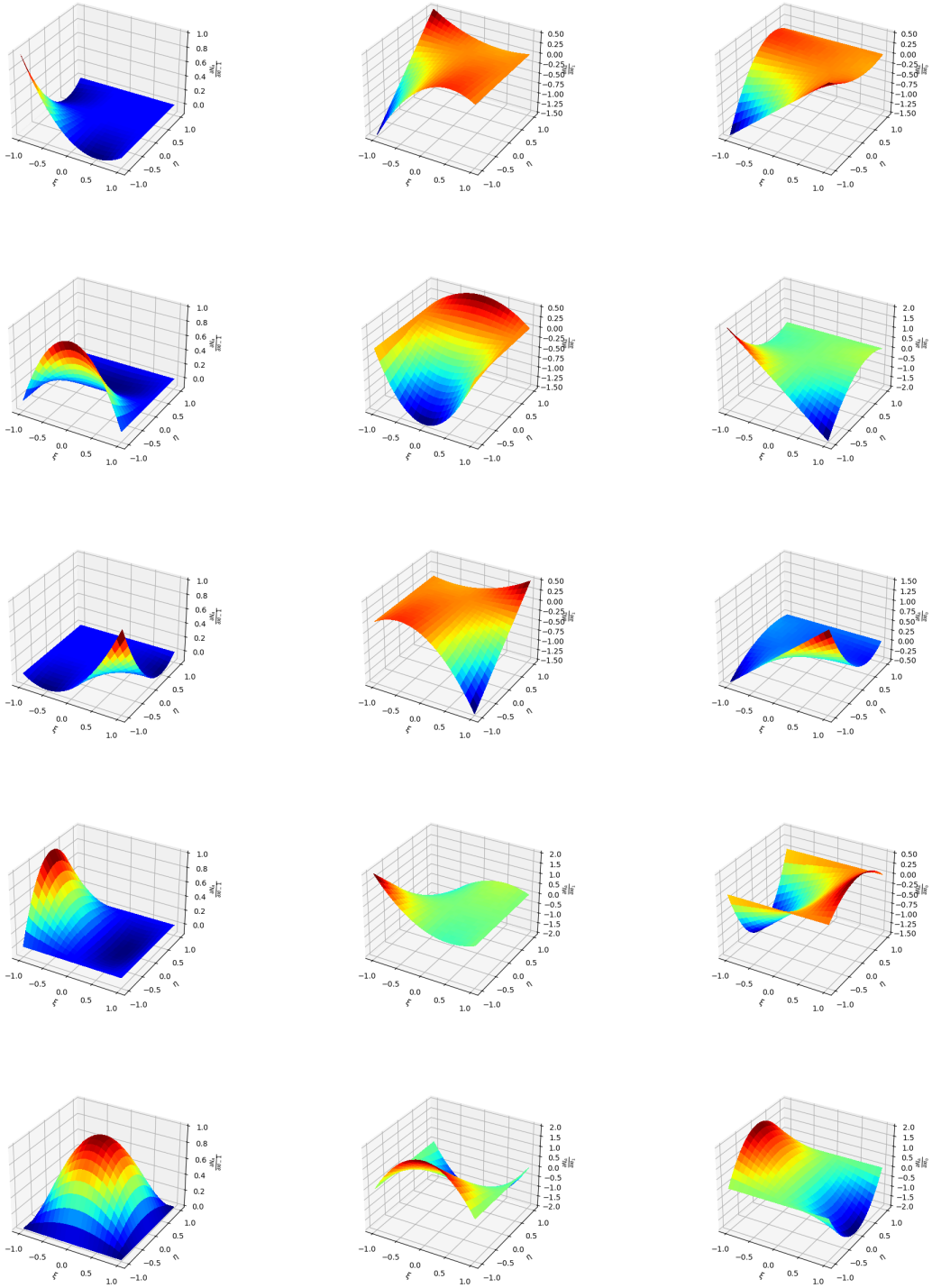
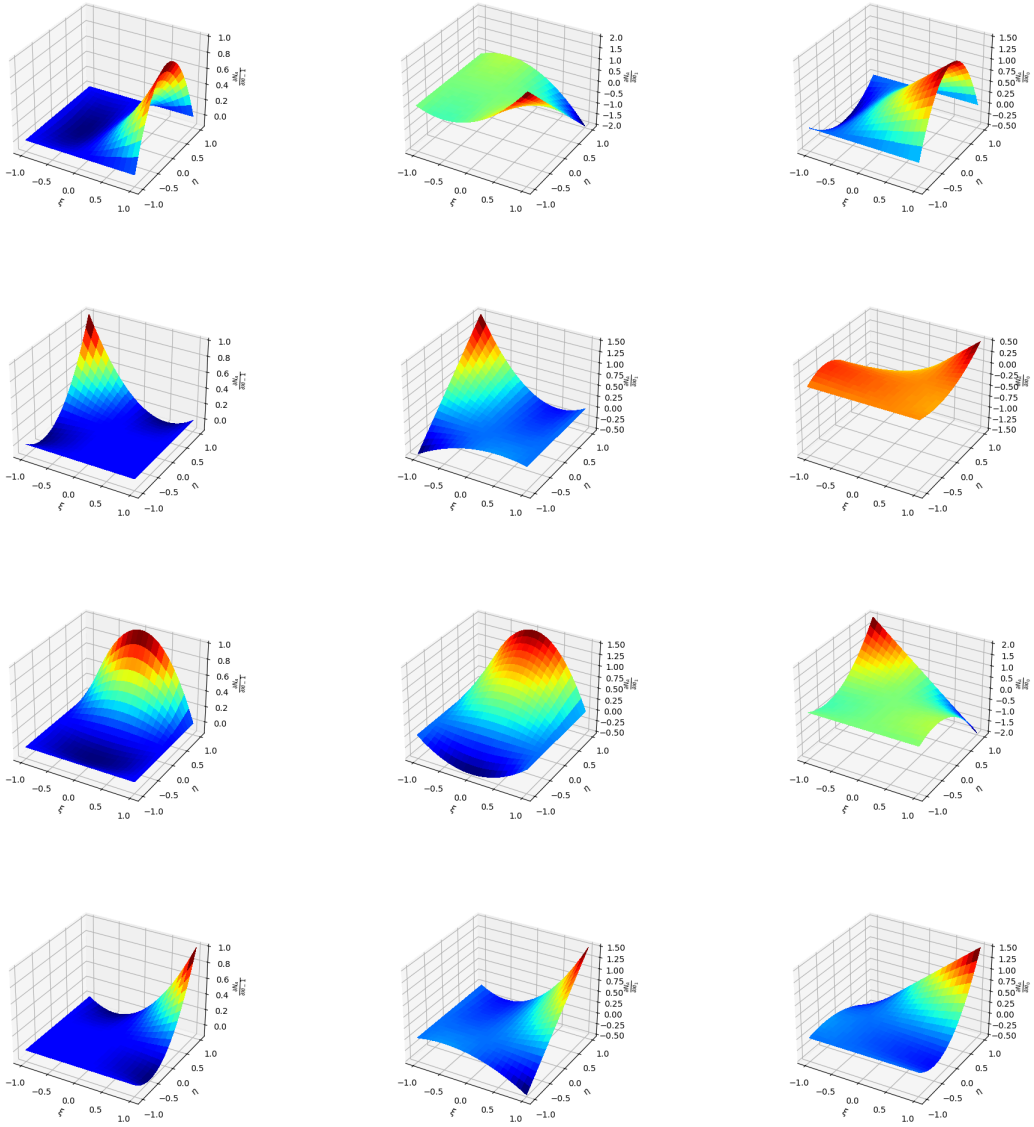
Problems 1 and 2 involved creating code to solve Equations 3/4 and Equation 6 from the handout. The output produced from this code is shown in Problems 3 and 4, and the code is attached to the LearningSuite submission for reference.

Problem 3

Below is a plot of the derivatives of the Lagrange basis functions of polynomial degree 2 on the interval [-1, 1] in one dimension. Note that they are all included on the same plot, but they are labeled using the legend. There are 3 derivatives because there are 3 basis functions.



Problem 4

This problem asked for the derivatives of the 2-dimensional Lagrange basis functions with respect to both Eta and Xi. The basis functions had degree 2 in both directions, so 9 total basis functions were generated. The resulting basis functions and their derivatives are included below. They are formatted so that each row corresponds to one basis function. The first column contains that basis function. The second column contains its partial derivative with respect to Eta, and the third column contains its partial derivative with respect to Xi.

Basis Function | Derivative w.r.t. Eta | Derivative w.r.t. Xi